

Signaling Road-Network Congestion: Aggregate V2V for Decentralized Traffic Control

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Abstract

Highway traffic jams are a congestion-control failure hiding in a transportation system: small local disturbances can push traffic past a critical density, after which stop-and-go waves persist long after the initial burst has disappeared. Prior self-regulating-car systems show that speed modulation can improve flow, but they still steer vehicles from global state broadcast by a cloud coordinator. This paper argues that highway congestion control can move into the traffic stream itself, and that the enabling question is one of communication abstraction. We envision ordinary vehicles as edge nodes that exchange only bounded, semantic, aggregate traffic summaries over a low-rate vehicle-to-vehicle substrate, build ego-centric digital twins of their local road state, and independently compute safety-filtered speed advisories. The abstraction we propose is neither high-bandwidth perception sharing nor command coordination, and it is distinct from the per-vehicle safety broadcast that existing connected-vehicle standards define; it is a *road-network congestion signal*, a small, stale-aware, lossy indicator of local traffic pressure that remains useful even when only a few vehicles participate. We specify this communication contract, describe the ego-centric digital twin and the model-based training approach that together support cross-topology adaptation, and outline the safety layer needed for mixed human traffic, grounding feasibility with Jetson-class perception measurements. The result is a decentralized, cloud-coordinator-free path toward traffic control that opens several interesting networking challenges.

1 Introduction

Traffic jams on free-flow highways are not only a capacity problem; they are a distributed stability problem. When local density crosses a critical threshold, small braking events, merge bursts, or overreactions can amplify into persistent stop-and-go waves even without a crash or a fixed bottleneck [3, 22]. The classical traffic-flow view captures this with the fundamental diagram: flow grows with density until a critical operating point, then collapses as interactions dominate [8, 15, 19]. Keeping density below that critical point, while holding speed high, keeps the whole stream on the productive branch of the diagram and maximizes sustained network throughput.

Small, well-timed local speed adjustments are enough to do this, which is what makes decentralized control a plausible lever on a network-level objective.

Fixed highway controllers are poorly matched to this failure mode. Ramp meters, variable speed limits, and centralized traffic-management systems act from fixed infrastructure and coarse network views [1, 17]. They are valuable where deployed, but they are expensive to install, unevenly available, and slow relative to local disturbances. A recent self-regulating-cars formulation showed that learned speed modulation can improve highway flow in simulation, but its controller observes global segment state and broadcasts recommendations through a cloud service [2]. That is a useful predecessor, but it leaves an open question: can traffic control move off both the roadside and the cloud, coordinated only by what vehicles can tell each other locally?

The timing makes this worth asking now. Connected and automated vehicles are moving from pilots to revenue service, and an evolving policy and regulatory environment is permitting more of them onto public roads, so the fraction of a highway stream that can be programmed is rising rather than hypothetical. The computation needed to estimate local traffic state is, separately, already widely available: robotaxis carry substantial on-board compute, modern smartphones can run the required perception and inference, and even human-driven vehicles increasingly ship with cameras, sensors, and on-board computers. What has been missing is not the computer but the coordination substrate and the regulatory room to use it, and both are now arriving.

The default trajectory, however, is that each automated vehicle optimizes only its own trip, so the network-control opportunity, the effect each car has on the flow around it, is left untapped. Highways make that opportunity controllable because their dominant problem is flow regulation rather than conflict arbitration. Surface-street intersections need a central arbiter because streams have mutually exclusive rights of way. Highways mostly have merges: vehicles interleave, and the control primitive is smooth speed and headway adjustment. That primitive already lives in the vehicle. The opportunity is to turn a sparse set of vehicles into mobile sensing, computing, and actuating nodes that regulate the flow from inside the stream.

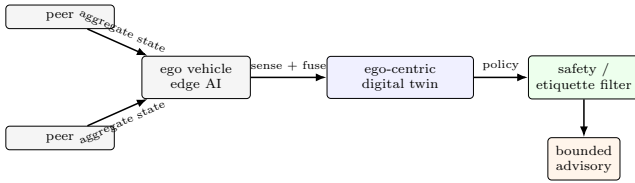


Figure 1: Self-regulating cars use low-rate aggregate V2V to improve each vehicle’s local observability. Each car independently maintains an ego-centric digital twin and emits only safety-filtered advisories; no roadside sensor, cloud coordinator, or command exchange is in the real-time loop.

Our key design choice is to make communication semantic and aggregate. Vehicles should not exchange raw video, persistent identities, direct commands, or lane-occupation plans. They should exchange compact traffic-state summaries: nearby AV and non-AV density, mean speed, queue pressure, downstream congestion, and merge pressure, from which each car builds its own ego-centric view. Control remains independent and local: each vehicle fuses onboard perception with aggregate vehicle-to-vehicle (V2V) state, updates a rolling digital twin, computes a speed and headway advisory, and passes that advisory through a safety layer before exposing it to a driver or automated interface.

This makes self-regulating cars a networking problem, and a specific one. Cars can already talk: connected-vehicle standards have them broadcast position and velocity for collision avoidance [6, 20]. The question is not whether a vehicle can run a neural detector, whether an optimizer can smooth a simulated merge, or whether a radio exists. It is what communication abstraction lets independently operated vehicles help a shared physical network without creating a new central planner. High-bandwidth perception sharing and V2V command exchange both entangle privacy, liability, and real-time authority, and, as we discuss, the per-vehicle safety broadcast that today’s connected-vehicle stack provides is not designed for flow regulation either. The abstraction we propose is closer to a congestion signal: small, stale-aware, lossy, local, and useful even when only a few vehicles participate.

This paper contributes a vision for this architecture. First, it reframes highway congestion control as a decentralized network-control problem coordinated by a low-rate aggregate-state substrate rather than by infrastructure or a cloud coordinator. Second, it specifies that substrate as a bounded, semantic, aggregate-state V2V contract, a road-network congestion signal, and distinguishes it from the per-vehicle broadcast of existing

connected-vehicle standards. Third, it identifies model-based training as a route to sample-efficient adaptation across road topologies, with the policy trained centrally and executed on purely local observation. Finally, it grounds the vision with measured Jetson-class pipeline feasibility and lays out the open networking questions needed to make such systems deployable.

2 Vision: A Road-Network Congestion Signal

The central claim is not that vehicles should form a fully cooperative swarm. It is that they should expose just enough local traffic state for nearby vehicles to make better independent decisions. This distinction matters for deployment. Command-style V2V coordination creates hard questions about authority, liability, and failure: who tells whom to slow down, which vehicle must occupy a lane, and what happens when messages are delayed? Self-regulating cars should avoid that contract. Each vehicle should remain the sole decision maker for its own motion, and network-level behavior should emerge from repeated local damping rather than from an agreed-upon plan.

Semantic, aggregate exchange. We therefore define V2V communication as a bounded semantic API. A sender contributes compact traffic-state summaries, and a receiver aggregates whatever it heard into its own local observation. The API is intentionally lower level than a plan and higher level than raw sensing. It gives a vehicle observability beyond its camera horizon without turning the network into a centralized planner, and without publishing anything that identifies or tracks an individual vehicle. Concretely, the contract admits and forbids two disjoint classes of state:

- **Exchanged:** nearby vehicle counts and density for both AVs and non-AVs, mean speed, lane distribution, local queue estimate, downstream congestion estimate, segment target speed, and merge pressure.
- **Never exchanged:** raw video, stable identities, direct acceleration or braking commands, joint lane plans, assignments of AVs to lanes, and coordinated roadblock behavior.

Relation to existing connected-vehicle messages. Vehicles can already broadcast to one another. Two deployed standards, the SAE J2735 Basic Safety Message (BSM) carried over Dedicated Short-Range Communications (DSRC), and the ETSI Cooperative Awareness Message (CAM) carried over Cellular Vehicle-to-Everything (C-V2X), have each vehicle emit its own position, speed, heading, and path history at up to ten hertz [6, 13, 20]. These messages are designed for collision avoidance: they

are per-vehicle and centered on the sender, describing where it is and where it is going. For flow regulation they are a poor fit: a receiver that wants a density or queue estimate must track many senders and fuse their trajectories, which reintroduces per-vehicle identity and linkage and the fusion cost the broadcast was meant to avoid. Our contract takes the complementary position. The unit of exchange is the receiver’s neighborhood aggregate, computed over whoever is currently in range, with no sender addressed, retained beyond the freshness window, or reconstructable from the aggregate. Because that aggregate is built from onboard perception as well as received summaries, it naturally covers non-AV (human-driven) vehicles too, not only cooperating AVs. The physical radio can be the same DSRC or C-V2X link; only the message semantics differ.

The contract. More formally, vehicle i maintains a local peer set

$$N_i(t) = \{j : \text{dist}(i, j) \leq R, t - \tau_j \leq T\},$$

where R is the aggregation radius, T is the freshness horizon, and τ_j is the receive time of the latest summary from peer j . The receiver reduces the summaries from $N_i(t)$ to a single aggregate observation $a_i(t) = A(\{b_j : j \in N_i(t)\})$, using count sums and confidence-weighted means over fields such as local count, mean speed, density, lane distribution, queue estimate, and merge pressure. The exact radio and aggregation operator can vary, but three semantics should not: expired peers are removed rather than extrapolated, unheard peers map to explicit neutral fields rather than zeros with hidden meaning, and $a_i(t)$ is advisory evidence rather than a command. In our prototype interface, this substrate is realized as low-rate broadcasts at 2 Hz, with a 2 s peer time-to-live (TTL) and a 150 m radius. This is a stand-in for a DSRC or C-V2X radio, not a claim about a specific one. If no peer is heard, the receiver falls back to neutral values and behaves as a local-only controller; if conflicting summaries arrive, it downweights low-confidence state rather than acting on any single message; and if bandwidth is scarce, it can trade rate or precision while still computing and filtering its own advisory.

Receiver-side semantics and privacy. The receiver-side rules are as important as the message format. A self-regulating car treats V2V state as an observation hint, not as authority. The local digital twin records when an aggregate was last refreshed, whether a field came from onboard sensing or V2V, and whether it was neutralized because no peers were heard. This lets the policy distinguish “I see no cooperating vehicles” from “the radio is stale”, and lets the safety layer reject advisories that depend on low-confidence state. It also keeps

the privacy surface small *by construction*: because the contract’s unit is a neighborhood aggregate, a participating vehicle need not publish camera frames, a persistent cross-session identifier, or its own trajectory for its neighbors to benefit from the density and speed information they collectively induce. Realizing that guarantee end to end, so that no raw per-vehicle position ever leaves the sender, is an open problem we return to in Section 6.

Why aggregate state is enough. Highway regulation does not require every vehicle to know the full network state, only early evidence that local flow is approaching a bad operating region: density rising near a merge, queues forming downstream, or speed variance increasing before a shockwave. This suffices because the control objective is physical and local to evaluate. The fundamental diagram relates flow to density, rising to a critical operating point and then collapsing as interactions dominate [8, 15, 19]; a regulator need only keep density below that threshold and then hold speed high and variance low, and both quantities are precisely what the aggregate signal conveys. A vehicle that damps its own speed variance when it sees rising local pressure removes energy from an incipient wave before it steepens, and when enough vehicles do this independently the stream is held near critical density with no shared plan. Onboard perception supplies immediate headway and leader and follower state, while aggregate V2V fills the blind spots beyond occlusions and around merges, so the observation is local but no longer myopic. The target is thus not consensus but local control under partial information, which is why the communication abstraction, rather than the optimizer, is the central design question.

Relationship to network control. The analogy to communication networks is direct: backpressure scheduling stabilizes queues by routing according to local pressure differences [24], and self-regulating cars seek an analogous pressure signal for physical traffic. The difference is that packets do not brake hard when startled and humans do not obey schedulers, which is why the communication contract must be paired with an ego-centric model of the scene and a safety layer that can reject any unsafe or obstructive advisory.

3 Architecture: Ego-Centric Digital Twin

The architecture has three loops: sensing and aggregate exchange, digital-twin update, and safety-filtered advisory. The real-time loop stays on the vehicle. Offline training may use centralized data and simulation, but deployment does not require roadside sensing, a cloud coordinator, or a globally consistent traffic map.

Sensing and fusion. Each vehicle estimates its immediate scene from onboard sensors: ego speed, acceleration, lane, headway, leader and follower gaps, relative speeds, local density, queue estimate, and distance to a merge. Aggregate V2V adds peer-derived summaries over a bounded radius. The combined observation is deliberately not a global state but an ego-local one, with a few semantic hints about what lies outside direct perception. The deployment stack mirrors the simulator’s observation contract rather than inventing a separate API, and logs every field with provenance (measured, derived, fallback-neutral, static, or sim-parity). Ego speed, for instance, is measured from GPS, held during short dropouts, and can be cross-checked against on-vehicle odometry; where several sensors observe one quantity the redundancy is reconciled and each field still carries its provenance tag. Leader gap and closing speed come from forward tracks, density from forward counts, rear-lane fields from neutral fallbacks until a rear camera is added, and nearby-vehicle fields from V2V when enabled. That provenance map is the basis for an observation-quality signal that both the policy and the safety layer can condition on.

The ego-centric digital twin. At runtime, each vehicle maintains a lightweight, continuously updated model of its local road state: the ego vehicle, nearby vehicles, lane structure, merge pressure, and short-horizon congestion risk. This is the digital twin. It is a compact tracker, not a city-scale simulator running in the car, and its job is observability. It lets the controller reason about partially observed density and downstream pressure instead of reacting only to the closest leader, and it lets the safety layer test an advisory against a short-horizon prediction before it is shown.

Centralized training and decentralized execution. The learning structure is centralized training with decentralized execution (CTDE). During training, a simulator or a logged fleet can expose global state, counterfactual rollouts, and gradients that are unavailable at deployment. During execution, the policy receives only onboard sensing and local-aggregate V2V fields. This split is the technical bridge from the centralized SRC predecessor [2] to a vehicle-native system: global information can teach the policy, but it is not part of the runtime control dependency, and no cloud coordinator sits in the loop.

Model-based training for sample efficiency. A purely model-free decentralized policy must relearn from many environment interactions each time it meets a new merge geometry, demand pattern, or compliance level. Model-based reinforcement learning is a well-established way to cut that cost: by learning a dynamics model and planning or generating rollouts within it, agents

reach comparable performance with substantially fewer real interactions [9, 10, 23]. We envision training the self-regulating-car policy in the same way, learning a dynamics model of the road network and using it to adapt the decentralized policy across topologies with few topology-specific interactions. This is a design position: given the sample-efficiency gains consistently reported for model-based RL, a learned road-network model is, in our view, the most promising route to cross-topology adaptation for a policy that must ultimately run on purely local observation.

Advisory control. The action space stays conservative: speed bins, headway targets, and cautious lane preferences rather than raw throttle, braking, or aggressive lane-change commands. The public action has four fields: desired speed bin (‘slow’, ‘nominal’, ‘fast’), desired headway bin (‘normal’, ‘larger’, ‘largest’), lane preference (‘keep’, ‘prefer left or right if safe’), and merge mode (‘normal’, ‘create gap’, ‘hold lane’). Prior work on self-regulating cars shows that discrete speed and headway bins of this kind are sufficient for a learned policy to improve flow and reduce hard braking [2]. A speed bin is decoded relative to the local segment-target-speed field and floored at a contextual minimum, and the decoded headway is fed back into the next observation. For human-in-the-loop deployment, the output is a bounded advisory in the spirit of prior advisory-autonomy work [4, 7, 29]; for automated vehicles, it can be passed to an existing longitudinal controller. In both cases the self-regulation layer nudges the vehicle toward network-friendly behavior without claiming full authority over the car.

Safety and etiquette filter. Every advisory passes through a local filter before it is shown or executed. The filter enforces posted speed limits, acceleration and deceleration bounds, minimum front and rear gaps, target-lane time-to-collision margins, a lane-change dwell time, a cap on lane changes per kilometer, and a limit on the braking an advisory may impose on target-lane followers. It also blocks low-speed behavior in uncongested conditions and synchronized all-lane slowdowns unless downstream congestion justifies them. Blocked or modified actions are surfaced through stable diagnostics: safety-masked action, etiquette block, follower-disruption block, and external override. This layer is not a polish detail; it is the reason a network objective can be exposed safely inside mixed human traffic, and it is what makes the action that is actually executed, rather than the raw policy proposal, the quantity that training and evaluation should measure.

Degraded operation. The edge design follows one rule: stale data is worse than missing data. If GPS goes

Table 1: Prototype measurements on Jetson Orin Nano. These numbers ground edge feasibility, not decentralized traffic-control effectiveness.

Component	Measured point	Use in system
YOLOv8n Ten- sorRT FP16 [18]	17.7 ms p50 detec- tion	Forward vehicle detections
Tracking + dis- tance	1.1 ms p50	Leader gap and closing speed
Observation + pol- icy	0.9 ms p50	Local observation and advisory
End-to-end com- pute	19.8 ms p50 / 20.2 ms p95 (syn- thetic bench)	Capture-to- advisory path
Simulated drives	34–38 ms p95	Real-time dash- cam + scripted GPS
Aggregate beacon	2 Hz, 2 s TTL, 150 m	Nearby AV sum- maries

stale, the observation builder holds the last valid speed for a short window rather than reporting a stop; if no V2V peers are heard, cooperation fields revert to explicit neutral values rather than silent zeros; if a display or logger stalls, the pipeline drops outputs rather than queuing an advisory that no longer matches the scene. These modes do not make the system complete, but they turn common in-car failures into explicit states rather than hidden dependencies, and they keep the driver-facing interface advisory-only, surfacing low confidence rather than hiding it inside the policy.

4 Feasibility: What We Can Ground Today

This paper is a vision, not a closed-loop traffic-control evaluation. The current evidence supports the edge-feasibility premise: a commodity embedded platform can run the perception, observation, and advisory loop needed to estimate local traffic state, and aggregate-state messages are small enough to avoid the bandwidth profile of raw perception sharing.

The latency and accuracy requirements are different from full autonomy. A self-regulating-car advisory does not need centimeter-accurate reconstruction of the scene; it needs stable estimates of headway, density, speed, and queue pressure at timescales relevant to human reaction and smooth longitudinal control. Our Jetson measurements suggest a plausible operating point for dashboard-grade or retrofit deployments on Jetson-class devices [16]. The prototype makes a latency-first choice: TensorRT YOLOv8n, lightweight tracking, and pinhole

or width-prior distance estimation rather than a dense monocular-depth network. That choice is swappable, but it grounds the claim that the observation and advisory loop can run comfortably below a 200 ms design budget. The end-to-end figures above are a synthetic benchmark and a real-detector simulated drive; an on-road camera run is future work.

The communication footprint is also modest. Raw or even compressed video is orders of magnitude larger than the semantic state needed here: the congestion signal carries counts, mean speed, lane distribution, queue estimate, and merge pressure, which at a few hertz is kilobits per second per vehicle and does not grow with scene complexity the way perception sharing would.

A simulated-drive harness gives a path from desk to road: it couples real dashcam footage with a scripted GPS profile and runs the full detector-to-advisory stack in real time, gating p95 latency, tick rate, GPS freshness, ego-speed error, and perception coverage. These gates do not prove that traffic control works, but they catch the systems failures (stale GPS, decode stalls, missing observations) that would otherwise contaminate a later traffic experiment.

Toward a pilot. What we establish here is edge feasibility, not a traffic-flow result. We plan to build toward the latter in a staged pilot: hardware-in-the-loop replay of many vehicles through the same stack under varying message loss, GPS delay, sensing range, and compliance; a microscopic-simulation study that replaces the predecessor’s global observation with the local observation and aggregate contract of Section 2, comparing local-only, local-plus-aggregate, and oracle global-state controllers under the same safety layer; and a closed-course study of whether advisories can be followed without uncomfortable braking or lane churn. The same pilot would exercise the model-based training of Section 3, testing whether a learned model cuts the interactions needed to adapt to an unseen topology.

5 Related Work

Sparse AVs as traffic actuators. Stern et al. showed that a single controlled vehicle can damp stop-and-go waves on a ring track [21], establishing the sparse-actuator premise. Flow and related mixed-autonomy RL work show that learned controllers can improve simulated traffic at low AV penetration [14, 27, 28]. CIRCLES/MegaVanderTest is the strongest field evidence that controlled vehicles can affect open-road traffic, but it uses purpose-built controlled vehicles and a centralized speed planner that assigns targets over cellular connectivity [5]. Our focus is different: a vehicle-native,

edge-executed design whose runtime communication is aggregate state rather than command dispatch.

Connected-vehicle messaging. Connected-vehicle standards let cars broadcast their own position, speed, and heading for collision avoidance: SAE J2735 Basic Safety Messages over DSRC and ETSI Cooperative Awareness Messages over C-V2X, at up to ten hertz [6, 13, 20]. These are per-vehicle and sender-centered, so recovering a density or queue estimate means tracking and fusing many senders. Our contract instead exchanges a receiver-side neighborhood aggregate in which no sender is addressed or reconstructable, reporting whether local flow is approaching a bad operating region rather than the position of any individual car.

Infrastructure-centric traffic control. Ramp metering, variable speed limits, adaptive signal control, and freeway management systems remain the dominant deployed tools [1, 17, 25, 26]. They are powerful where sensors and actuators exist, but they depend on fixed infrastructure and centralized operational visibility. Self-regulating cars instead treat vehicles as mobile actuators that can be introduced incrementally.

Advisory systems. Human-compatible advisory systems show how network objectives can be translated into driver-facing recommendations [4, 7, 29]. This matters because early deployment will be mixed: some vehicles may be automated, some may show dashboard advisories, and many will ignore the system. The open systems question is therefore not only control optimality, but compliance, incentives, and fairness under partial adoption.

Edge AI and networked sensing. Embedded perception models and Jetson-class hardware make on-vehicle traffic-state estimation increasingly practical [16, 18, 30]. Cloud video analytics demonstrates the value of centralized visual processing but also the cost of moving raw data off the edge [11, 12]. Our communication model takes the opposite position: infer locally, exchange only semantic aggregate state, and keep safety-critical control on the vehicle.

6 Discussion: Open Networking Questions

Treating self-regulating cars as a communication-abstraction problem turns their hardest questions into networking questions. Several are central.

What is the right semantic API? The contract of Section 2 is a starting point. The community needs principled positions on rate, precision, aggregation radius, freshness horizon, and trust: how stale a signal may be before it misleads, how coarse before it stops helping, and how a receiver should weight summaries of unknown

provenance. That points to work on adaptive beacon rates, confidence-weighted aggregation, and semantic compression under packet loss.

Can the aggregate be computed privately? Our contract keeps the semantic unit anonymous: the receiver consumes a neighborhood aggregate, not a per-vehicle record. But a naive realization still has each sender emit its own position, as a Basic Safety Message already does. The open problem is to produce the aggregate without any raw per-vehicle trajectory leaving the vehicle, through in-network or gossip-style aggregation, randomized-response or differentially private counts, and neighborhood summaries that are useful for a flow regulator yet carry no linkable identity.

How robust is the system to partial deployment?

The relevant adoption unit is not market share but the fraction of vehicles present in a specific traffic stream. Evaluation should sweep penetration, compliance, message loss, sensing range and noise, and demand bursts across a topology ladder (ring roads, straight highways, merges, and inverted trees), with metrics spanning throughput, jam fraction, and delay alongside safety and passability measures such as hard braking, collisions, and follower delay, so that a network gain is never bought by making human traffic less safe or less passable.

How should safety be verified? A local policy can optimize a network objective and still be unacceptable if it surprises human drivers. The safety layer should be specified, tested, and audited separately from the learned controller, with explicit checks for stale state, follower disruption, low-speed obstruction, and unsafe merges. This separation can become a systems requirement: the execution contract can report how often a suggestion was masked, blocked for etiquette, blocked for follower disruption, or overridden, giving evidence that a network objective is being achieved safely.

Who benefits, and who decides? Infrastructure-free control is most compelling where roadside upgrades are unlikely, but a voluntary advisory system can also shift delay across drivers, lanes, or vehicle classes. A deployable system therefore needs measurement and incentives that make the network benefit visible and its distribution accountable.

As vehicles become networked computers, congestion control need not remain a roadside service. The controller and the radio already exist; the open question is how to design the communication substrate so that each vehicle sees enough of the local traffic state to help the network while still acting independently, safely, privately, and without a cloud coordinator. We think this is a productive new problem space for the networking community.

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